

Gravitational waves scattering off a spherical thin shell

Summary: junction conditions, reflectivity, quasinormal modes, stability, and the two polarizations

Final project, gravitational-wave course

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Abstract

A thin spherical shell of matter at radius $R > 2M$ separates a flat interior from a Schwarzschild exterior of mass M . It is the simplest horizonless compact object. We compute, for odd (axial) and even (polar) gravitational perturbations, the reflection and transmission coefficients $\mathcal{R}(\omega)$ and $\mathcal{T}(\omega)$ and the quasinormal-mode (QNM) spectrum. The first step is to find a background that can be static at all. A domain wall cannot be; a fluid shell with a specific positive pressure can. For odd parity the shell then turns out to be a repulsive δ -function in the wave equation. For even parity it is a flux-conserving 2×2 transfer matrix. The shell is opaque at low frequency and transparent at high frequency, and it approaches the black-hole reflectivity as $R \rightarrow 2M$. Its ringing is completely different from a black hole's, with long-lived trapped modes for $R < 3M$. One even-parity mode of the shell's own matter grows exponentially, so the static shell is unstable, in agreement with Pitre, Schneider and Poisson (2026). Finally, we translate the two parities into the polarizations h_+ and $h_\times$. Parity is conserved, but the object is a birefringent mirror: it reflects the two parities with different phases, so a general wave changes polarization. The effect is universal at long wavelengths and complete at the even-only matter resonance. Units: $G = c = 1$, time dependence $e^{-i\omega t}$.

1 The problem and the first obstacle: is there a static shell?

Inside the shell, Birkhoff's theorem and regularity at $r = 0$ force flat space: $ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2$. Outside, $ds^2 = -f dt^2 + f^{-1} dr^2 + r^2 d\Omega^2$ with $f = 1 - 2M/r$. The shell carries a surface stress tensor $S_{ab} = \sigma u_a u_b + p (h_{ab} + u_a u_b)$. Gluing the two geometries is governed by the Israel junction conditions [4, 6]:

$$[h_{ab}] = 0, \quad [K_{ab}] - h_{ab}[K] = -8\pi S_{ab}, \quad [X] \equiv X_+ - X_-, \quad (1)$$

where h_{ab} is the induced metric and K_{ab} the extrinsic curvature. For a spherical shell $r = R(\tau)$ they give the mass formula and the equation of motion (Ipser & Sikivie [5], Eqs. 3.7–3.9):

$$\sigma = \frac{\sqrt{1 + \dot{R}^2} - \sqrt{f(R) + \dot{R}^2}}{4\pi R}, \quad (\alpha + \beta)\ddot{R} = -\frac{\alpha M}{R^2} + \frac{2p}{\sigma} \frac{\alpha\beta(\alpha + \beta)}{R}, \quad (2)$$

with $\alpha = \sqrt{1 + \dot{R}^2}$ and $\beta = \sqrt{f + \dot{R}^2}$.

Idea. A monochromatic scattering problem ($e^{-i\omega t}$) needs a time-independent background. Before any wave calculation we must therefore ask whether the shell can sit still.

Gravity always pulls inward ($-\alpha M/R^2$). Pressure pushes outward only if $p > 0$. Setting $\dot{R} = \ddot{R} = 0$ gives a unique answer:

$$\frac{p}{\sigma} = \kappa(R) = \frac{1 - \sqrt{f_R}}{4\sqrt{f_R}} > 0, \quad \sigma = \frac{1 - \sqrt{f_R}}{4\pi R}. \quad (3)$$

This leaves three candidate models (a de Sitter interior, the gravastar [12], is not treated):

- **Domain wall** ($p = -\sigma$, $S_{ab} = -\sigma h_{ab}$). $\ddot{R} < 0$ always, so it collapses to a black hole [5]. It can only be *frozen* at its turning point (*model A*), where $\ddot{R} = -(1 + 3\sqrt{f_R})/2R$.
- **Dust** ($p = 0$). It is not static either: $(\alpha + \beta)\ddot{R} = -\alpha M/R^2 < 0$.
- **Static perfect fluid** with $p = \kappa(R)\sigma$ (*model C*, the primary model). The pressure ratio is fixed by the compactness, not free. The energy condition $p \leq \sigma$ requires $R \geq 25M/12$. For the perturbed shell we need an equation of state, $\delta p = \Gamma \frac{p}{\sigma + p} \delta\sigma$ with $\Gamma = 2$, as in [15].

The interior clock runs slower: $dT = \sqrt{f_R} dt$. A wave of frequency ω outside therefore has frequency $\nu = \omega/\sqrt{f_R}$ inside.

2 Waves on each side and the junction conditions

Idea. Away from the shell everything is textbook. Outside, the Regge–Wheeler/Zerilli equations hold; inside, the flat wave equation. *All* the physics of the shell is in how the two solutions are glued at $r = R$.

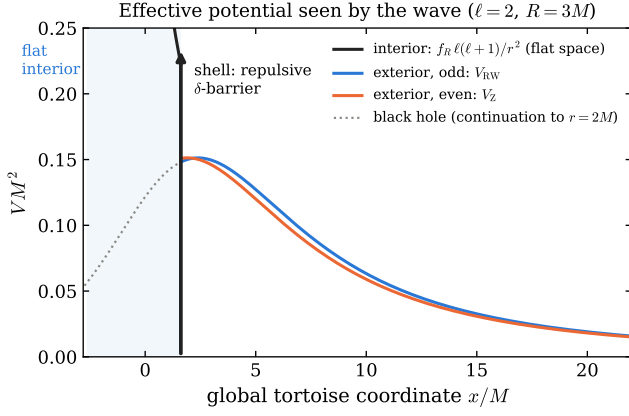


Figure 1: Potential seen by the wave. Flat interior (centrifugal barrier), shell (δ -barrier, arrow) and exterior RW/Zerilli barrier.

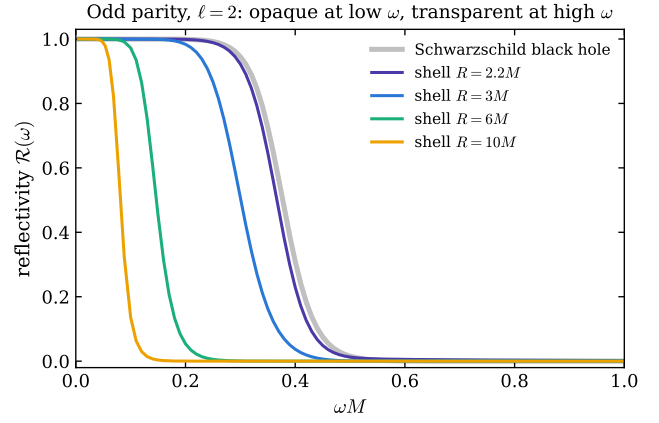


Figure 2: Reflectivity for odd parity, $\ell = 2$. Compact shells approach the black-hole curve.

We use the gauge-invariant master functions of Martel and Poisson [7]: Cunningham–Price–Moncrief for odd parity and Zerilli–Moncrief for even parity. Outside, $\Psi_{,r_*r_*} + [\omega^2 - V]\Psi = 0$, with $V_{\text{RW}} = f[\ell(\ell+1)/r^2 - 6M/r^3]$ [1] or the Zerilli potential V_Z [2]. Inside, $\Psi'' + [\nu^2 - \ell(\ell+1)/r^2]\Psi = 0$. Its regular solution is the Riccati–Bessel function $\hat{j}_\ell(\nu r)$. In a global tortoise coordinate x ($x = r_*$ outside, $dx = dr/\sqrt{f_R}$ inside) the whole problem is a 1D wave equation with the potential of Fig. 1.

The linearized Israel equations (1) were derived from first principles. On each side the metric perturbation is written in its own Regge–Wheeler gauge. Because those gauges are not adapted to the shell, the shell’s position is described by perturbed embeddings, with radial displacements $\xi_\pm$ that differ on the two sides. The metric perturbations are then expressed through the master functions. The derivation was done symbolically (SymPy) and verified. The results are the following.

Odd parity: a δ -barrier. Odd perturbations do not move the shell or its matter. The background term $[K]^{(0)}$, which one might expect to depend on the pressure (or on $\dot{R}$ for a wall), cancels exactly against the pressure term. What remains is

$$\Psi_+ = \Psi_-, \quad \left[\frac{d\Psi}{dx} \right] = \Delta \Psi, \quad \Delta = \frac{\sqrt{f_R}(1 - \sqrt{f_R})}{R} = 4\pi\sigma\sqrt{f_R} > 0. \quad (4)$$

The shell acts on axial waves as a repulsive $\delta(x - x_R)$ potential. Its strength does not depend on ω , on the equation of state, or on whether the shell is a wall or a fluid. The same condition was recovered independently by letting a smooth anisotropic shell become thin; the mismatch vanishes linearly in the thickness. It also agrees with [15] and with the matching conditions of [12].

Even parity: a 2×2 transfer matrix. Polar perturbations move the shell radially and make its fluid oscillate. After eliminating the shell variables ($\xi_\pm$, $\delta\sigma$, velocity) one gets

$$\begin{pmatrix} \Psi_+ \\ \Psi'_+ \end{pmatrix}_R = \mathsf{T}(\omega; \ell, R/M, \Gamma) \begin{pmatrix} \Psi_- \\ \Psi'_- \end{pmatrix}_R, \quad \det \mathsf{T} = f_R^{-1/2} \text{ (exactly)}. \quad (5)$$

Ψ itself jumps across the shell, and T depends on frequency, with poles on the matter scale $\omega \sim (M/R^3)^{1/2}$. The determinant identity was not imposed; it is what guarantees energy conservation, and it holds to machine precision.

3 Scattering: how to define $\mathcal{R}$ and $\mathcal{T}$, and what they are

Idea. Nothing is absorbed, so the *closed* system reflects everything: $|B_{\text{out}}/B_{\text{in}}| = 1$. To ask how much the object reflects, we *open* the interior. A wave that crosses the shell is let go (only the ingoing Hankel wave $\hat{h}_\ell^{\text{in}}(\nu r)$ inside), exactly as a horizon lets it go for a black hole.

Outside, $\Psi \rightarrow B_{\text{in}}e^{-i\omega r_*} + B_{\text{out}}e^{i\omega r_*}$, and we define $\mathcal{R} = |B_{\text{out}}/B_{\text{in}}|^2$ and $\mathcal{T} = 1/|B_{\text{in}}|^2$. Flux conservation through the junction then implies $\mathcal{R} + \mathcal{T} = 1$. Figures 2–3 show four regimes:

- **Low frequency:** $\mathcal{R} \rightarrow 1$. Long waves are reflected, as for any compact object [14].
- **High frequency:** the potentials become negligible and only the δ -barrier is left. Then $\mathcal{R} \simeq \Delta^2/(4\omega^2 + \Delta^2) \rightarrow 0$: *the thin shell is transparent*. Even parity (model C) follows the same law, because the fluid’s inertia no longer matters (Fig. 3).
- **Compact shells, $R \rightarrow 2M$:** $\Delta \rightarrow 0$ and the shell sits deep in the redshifted throat, so $\mathcal{R} \rightarrow \mathcal{R}_{\text{BH}}$ (Vishveshwara [3]). The approach is slow, $\propto (R - 2M)^{1/2}$.
- **Dilute shells:** the transition to transparency happens at $\omega R \sim \ell$, set by the flat-space centrifugal barrier. Even and odd reflectivities differ at intermediate frequencies. The shell breaks the isospectrality of Schwarzschild.

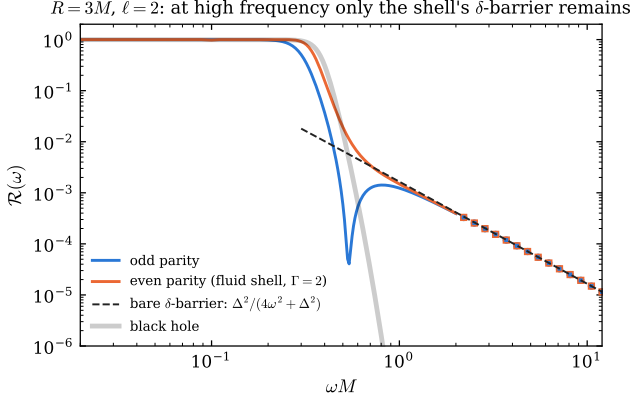


Figure 3: Odd and even ($\Gamma = 2$) reflectivity at $R = 3M$ on log axes, with the δ -barrier law. Dots: points computed up to $\omega M = 12$.

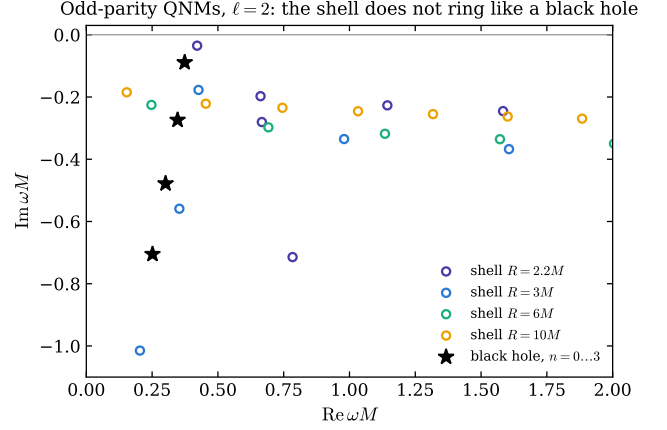


Figure 4: Odd QNMs ($\ell = 2$) for four compactnesses, against the Schwarzschild modes (stars).

4 Quasinormal modes: how the shell rings

QNMs are the complex ω at which the regular interior solution, carried through the junction, is purely outgoing at infinity. They are found as zeros of a Wronskian. The outgoing solution comes from a Leaver-type continued fraction [8, 12] and independently from inward integration along a complex ray [14]. Three families appear (Figs. 4, 5):

1. **Wave modes** of the interior cavity, with $\omega \sim 1/R$. For dilute shells $\text{Re}(\omega R) \approx n\pi$ and $\text{Im}(\omega R) \approx -\frac{1}{2} \ln[2\ell(\ell+1)R/M]$ [15]. The cavity leaks through a weakly reflecting wall, so the damping is large and grows only logarithmically with R .
2. **Trapped modes** for $R < 3M$, between the flat interior and the exterior photon-sphere barrier. They are damped only by tunnelling and become extremely long-lived as $R \rightarrow 2M$: $\text{Im} \omega M \approx -10^{-6}$ at $R = 2.01M$. These are the modes behind gravitational-wave *echoes* [13]. *None of them tends to a Schwarzschild QNM*, even though the reflectivity does tend to the black-hole one.
3. **Matter modes** (even parity, model C only): oscillations of the shell itself, four per ℓ , with $\omega \sim (M/R^3)^{1/2}$. One of them is purely imaginary with $\text{Im} \omega > 0$.

5 The static shell is unstable

Idea. The only static member of the family, model C, should be stable if scattering off it is to describe something physical. The growing matter mode shows that it is not. It is then essential to compare time scales.

At $M/R = 0.2$, $\ell = 2$, $\Gamma = 2$ the growing mode is $\omega(R^3/M)^{1/2} = 0.6077i$, and the oscillating pair is $1.2671 - 0.0011i$. These values agree with Pitre, Schneider and Poisson [15] at every compactness (Fig. 5), although their method (retarded-time slicing, spectral solver) is completely different. As $M/R \rightarrow 0$ they tend to the oscillations of a *Newtonian* self-gravitating fluid shell. We derived that problem from scratch; it gives $\zeta^4 - S\zeta^2 + P = 0$ with $\zeta^2 = \omega^2 R^3/M$, and for $\ell = 2$ the roots are 1.5050 and $0.5147i$. The instability is therefore Newtonian in origin: the tangential pressure that holds the shell up makes $\ell \geq 2$ deformations run away. The same derivation shows that the constant term of the quartic in [15], Eq. (7.6), contains a misprint; their closed form, Eq. (7.8), is correct.

Consequence for the models (Fig. 6). The fluid shell grows on $\sim 1.6(R^3/M)^{1/2}$, which is much longer than the wave periods ($\sim R$). Scattering off model C is therefore a meaningful adiabatic idealization. The domain wall collapses on $\omega_{\text{dyn}}^{-1} \approx R/\sqrt{2}$, the light-crossing time, which is comparable to the wave periods. For the frozen wall (model A) the even-parity junction system is overdetermined and inconsistent (8 equations, 6 unknowns). Which two Israel equations are dropped is a free choice, and different choices change $\mathcal{R}$ at $O(1)$, from total reflection to the δ -law. **The even-parity results of model A are not predictive;** its odd sector is identical to model C's.

6 The two polarizations: h_+ , $h_\times$ and the two parities

Idea. A detector measures h_+ and $h_\times$, not Ψ_{even} and Ψ_{odd} . Is “even = plus”, can reflection turn $h_\times$ into h_+ , and does one parity matter more?

Dictionary. Far away [7] (Eqs. 6.13–6.16),

$$h_+ - ih_\times = \frac{1}{2r} \sum_{\ell m} \sqrt{\frac{(\ell+2)!}{(\ell-2)!}} (\Psi_{\text{even}}^{\ell m} - i\Psi_{\text{odd}}^{\ell m}) {}_{-2}Y^{\ell m}, \quad \dot{E} = \frac{1}{64\pi} \sum_{\ell m} \frac{(\ell+2)!}{(\ell-2)!} (|\dot{\Psi}_{\text{even}}^{\ell m}|^2 + |\dot{\Psi}_{\text{odd}}^{\ell m}|^2). \quad (6)$$

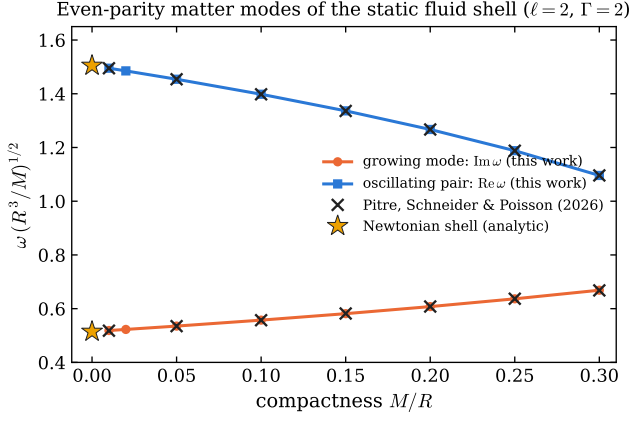


Figure 5: Matter modes of the fluid shell vs compactness: growing mode (orange) and oscillating pair (blue). Crosses: [15]. Stars: Newtonian shell.

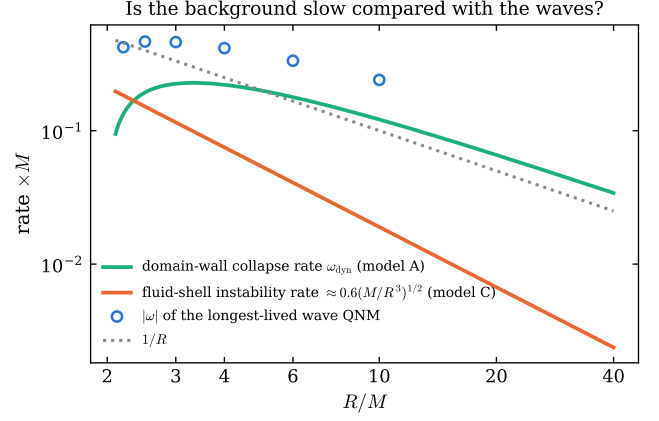


Figure 6: Background rates vs the least-damped wave QNM. The wall collapses as fast as the waves oscillate; the fluid instability is slow.

Each parity produces both polarizations. The odd field is the even one rotated by 45° in polarization (the factor $-i$): these are E - and B -type patterns. Only for $m = 0$ is even $= h_+$ and odd $= h_\times$ literally. Both parities carry energy with the same weight.

Selection rule and birefringence. The background is spherically symmetric and parity invariant, so even and odd never mix. An axisymmetric pure- $h_\times$ wave comes back as pure $h_\times$. For a single (ℓ, m) the object acts on the (E, B) components as the Jones matrix $\text{diag}(S_{\text{even}}, S_{\text{odd}})$, i.e. as a birefringent mirror. For a wave with equal E and B content, a fraction $\sin^2(\Phi/2)$ of the reflected power comes back with the orthogonal polarization, with retardance $\Phi = \arg(S_{\text{even}}/S_{\text{odd}})$. The same fraction gives the helicity-flip probability.

- *Black hole.* The Chandrasekhar map gives exactly $S_{\text{even}}/S_{\text{odd}} = (\lambda + 12iM\omega)/(\lambda - 12iM\omega)$ with $\lambda = (\ell - 1)\ell(\ell + 1)(\ell + 2)$ [16]. Isospectral ($|S|$ equal), but the phase differs: $\Phi = 21^\circ$ at the $\ell = 2$ QNM frequency. The numerical ratio agrees to 5×10^{-8} . For a plane wave the helicity-reversing cross section, $\sigma_{\text{flip}} = \frac{\pi}{\omega^2} \sum (2\ell + 1) |S_{\text{even}} - S_{\text{odd}}|^2 / 4$, tends to $144\pi M^2 \sum (2\ell + 1) / \lambda^2 = 4\pi M^2 / 3$ as $\omega \rightarrow 0$. This is exactly the helicity-reversing part of the classic long-wavelength cross section [17, 18].
- *Shells.* $|S_{\text{even}}| = |S_{\text{odd}}| = 1$ (to 10^{-15}): both parities return all their energy. At low frequency the retardance and $\sigma_{\text{flip}} = 4\pi M^2 / 3$ are the black-hole ones, a universal effect of the exterior field. At the even-only matter resonance ($\omega M = 0.197$ for $R = 3M$, 0.089 for $6M$) the phase of S_{even} winds by 2π and the conversion becomes complete in a narrow band (Fig. 7). Elsewhere it is small: a few per cent, and up to 19% between the even and odd trapped resonances at $R = 2.2M$. With an absorbing interior the even parity is also reflected more strongly, by up to an order of magnitude near the barrier top.

Waveforms. The same incident pulse, carried once by Ψ_{even} and once by Ψ_{odd} , gives nearly identical prompt reflections (Fig. 8). The even channel then differs: dephased trapped ringing, a long ringing at the matter frequency, and the unstable mode. We added the latter causally from the residue of S_{even} at its pole (found at $0.2442921i$ for $R = 2.2M$, the known unstable frequency). It is weakly excited but overtakes the signal after 74 – $440M$ ($R = 2.2$ – $6M$). The odd synthesis agrees with an independent time-domain evolution to $< 1\%$.

Which parity matters? In energy neither: equal weights, equal content in a plane wave, equal reflected energy. In physics the even parity is richer: it alone couples to the shell’s matter (resonances, instability). It also dominates the emission of slow sources; the current quadrupole is suppressed by $\frac{1}{36}(\delta m/m)^2 v^2$ in power for a binary [19]. The odd parity is the simple, universal one, a δ -barrier.

7 How we know it is right

test	result
energy balance $ \mathcal{R} + \mathcal{T} - 1 $ (odd, even model C, $\omega M \leq 2$)	$\lesssim 6 \times 10^{-10}$
transfer matrix vs direct shooting: $\mathcal{R}, \mathcal{T}$ / 397 QNMs	$\lesssim 10^{-9}$ / median 4×10^{-12} , worst 6×10^{-5}
Schwarzschild $\ell = 2$, $n = 0$ (Leaver); WKB orders $1/3/6$	$0.3736716844 - 0.0889623157i$ [11]; 7%/0.15%/0.02% [9, 10]
odd junction from a thick-shell limit; $\det T \sqrt{f_R} = 1$ (even)	converges $\propto$ thickness; 10^{-12}
time-domain ringdown vs QNM, $R = 2.2M$ / black hole	$0.4215 - 0.0352i$ vs $0.4211 - 0.0350i$ / $0.3731 - 0.0885i$ vs $0.3737 - 0.0890i$
BH: $S_{\text{even}}/S_{\text{odd}}$ vs Chandrasekhar phase; $\sigma_{\text{flip}}(\omega \rightarrow 0)$	5×10^{-8} ; $4.188M^2$ vs $4\pi M^2/3 = 4.1888M^2$
odd waveform: frequency-domain synthesis vs time-domain evolution	$< 1\%$ of the peak

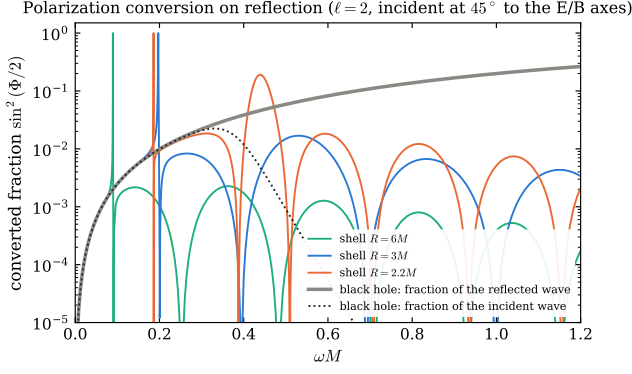


Figure 7: Fraction of the reflected power returning with the orthogonal polarization ($\ell = 2$, equal E and B content). Complete conversion at the matter resonances; the black hole for comparison.

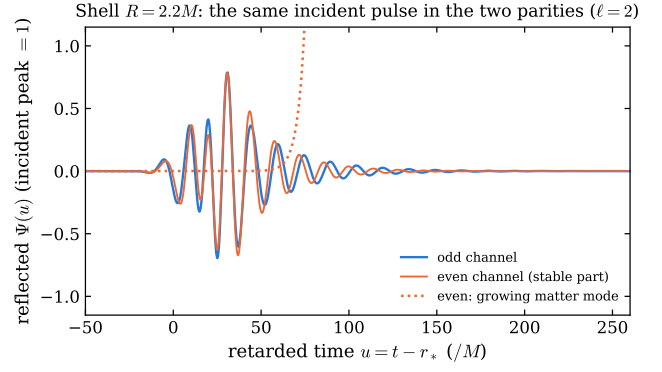


Figure 8: Reflected waveform of the same pulse in the two parities, $R = 2.2M$. Dotted: the growing matter mode of the even channel.

8 Conclusions

1. A thin shell with a flat interior is static only if it has positive surface pressure, $p = \sigma(1 - \sqrt{f_R})/4\sqrt{f_R}$. Domain walls and dust shells collapse.
2. For axial waves the shell is a repulsive δ -potential, $\Delta = \sqrt{f_R}(1 - \sqrt{f_R})/R$. For polar waves it is a frequency-dependent, flux-conserving 2×2 transfer matrix, which also contains the dynamics of the shell's matter.
3. $\mathcal{R}(\omega)$ goes from total reflection at low frequency to transparency at high frequency ($\mathcal{R} \propto \omega^{-2}$). It approaches the black-hole reflectivity as $R \rightarrow 2M$, while the QNM spectrum does not approach the black-hole one. Compact shells trap long-lived modes, which produce echoes.
4. The static shell is dynamically unstable through a Newtonian-origin matter mode. The instability is slow compared with the waves, so the scattering results for the fluid shell are physically meaningful. The frozen domain wall is not a controlled approximation in the even sector.
5. Each parity produces both h_+ and $h_\times$. The odd field is the even one rotated by 45° in polarization, and the two carry energy with equal weight. Parity is exactly conserved, so a pure-parity wave (e.g. an axisymmetric $h_\times$ wave) never turns into the other polarization. Mixed waves do change polarization, because $S_{\text{even}} \neq S_{\text{odd}}$: universally at long wavelengths ($\sigma_{\text{flip}} = 4\pi M^2/3$), and completely at the even-only matter resonance.
6. Neither parity dominates the energy of a scattered wave. The even parity carries all the coupling to the shell's matter (resonances, instability) and dominates the emission of slow sources. The odd parity is simpler and universal.

Status. This summary and the accompanying written material are final: no new calculations or results will be added. Only the accompanying presentation ([3.presentation.html](#)) may still be revised.

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